

Cursor hackathon

East Ethiopia

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1. System Overview

The proposed system is a **Trusted Hyperlocal Commerce Platform** designed for the Ethiopian market. The platform connects **verified merchants and service providers with nearby buyers**, while ensuring secure transactions through **M-Pesa-based escrow payments** and **platform-managed delivery logistics**.

Unlike open marketplaces where anyone can register and sell products, this system operates under a **verified merchant model** where merchant accounts are created and approved by the platform administrator. This ensures trust, reduces fraud, and maintains high-quality service standards.

The system integrates three major infrastructures into a single platform:

1. **Digital Marketplace Infrastructure** – enabling discovery and promotion of products and services.
2. **Secure Transaction Infrastructure** – ensuring payment safety using escrow with M-Pesa.
3. **Logistics and Delivery Infrastructure** – coordinating pickup and delivery through the platform's delivery network.

Through these integrated services, the platform aims to solve fundamental challenges in local commerce such as **trust, discoverability, secure payments, and reliable delivery**.

2. Key Stakeholders

The system involves three primary actors:

Buyers (Customers)

Individuals who use the platform to discover nearby merchants and service providers, purchase products, request services, track deliveries, and evaluate merchants through ratings and reviews.

Merchants / Service Providers

Businesses or individuals offering products or services on the platform. Their accounts are created and verified by the system administrator to ensure authenticity and quality.

System Owner (Platform Administrator)

The entity responsible for operating the platform, managing merchant verification, controlling escrow transactions, overseeing delivery logistics, resolving disputes, and maintaining system integrity.

3. Core Objectives of the Platform

The system aims to achieve the following objectives:

- Provide a **trusted marketplace** where buyers can confidently purchase products or hire services.
- Enable **small and medium merchants** to reach more customers through digital visibility.
- Reduce fraud through **verified merchant onboarding and escrow-based payments**.
- Provide **reliable delivery coordination** through a platform-controlled logistics network.
- Build a **reputation-driven ecosystem** through ratings, reviews, and merchant trust scores.
- Generate sustainable revenue for the system owner through **merchant promotion and service commissions**.

4. System Functional Modules

The platform is composed of several major functional modules.

4.1 Buyer Module

The Buyer Module provides the tools necessary for customers to discover products and services and perform secure transactions.

Account Management

Buyers can:

- Register and create a user account
- Link their **M-Pesa wallet**
- Manage delivery addresses
- View order history

Marketplace Discovery

Buyers can search and explore the marketplace using filters such as:

- Product category
- Service category
- Merchant rating
- Distance from location
- Price range
- Popularity

The system uses **location-based filtering** to prioritize nearby merchants and reduce delivery times.

Product Viewing

Buyers can view detailed product information including:

- product images
- descriptions
- price
- merchant details
- merchant rating
- number of completed orders
- estimated delivery time

Service Discovery

For service providers, buyers can view:

- service descriptions
- service areas
- pricing range
- portfolio or examples
- customer reviews
- provider ratings

Order Placement

Buyers can place orders for products through the platform.

The system calculates:

Product price

- Delivery fee
- Platform service fee (if applicable)

Escrow Payment via M-Pesa

When a buyer confirms an order:

1. The buyer pays using **M-Pesa**.
2. The payment is transferred to the **platform escrow account**.
3. The merchant does not receive the payment immediately.

This ensures that the buyer is protected until the order is successfully delivered.

Order Tracking

Buyers can track orders in real time with status updates such as:

- Order placed
- Merchant confirmed
- Pickup scheduled
- Item collected
- In transit
- Delivered

Review and Rating System

After delivery, buyers can rate the merchant using:

- star rating
- written review
- optional images

Reviews contribute to the merchant's **reputation score**.

4.2 Merchant Module

Merchants use the platform to manage their products, orders, and promotions.

Merchant accounts are **created by the platform administrator after verification**.

Merchant Profile

Each merchant has a profile that includes:

- business name
- business description
- location
- operating hours
- verification badge
- merchant rating
- product categories

Product Management

Merchants can:

- add new products
- upload images
- update product prices
- manage inventory
- categorize items

Order Management

Merchants can view incoming orders and perform actions such as:

- confirm order
- prepare product for pickup

- communicate with the platform if issues arise

Merchants do not handle delivery directly; pickup is managed by the platform logistics system.

Merchant Analytics

Merchants have access to performance insights including:

- total sales
- most popular products
- customer feedback
- order completion rates

Merchant Promotion (Boost System)

Merchants can increase visibility by paying the platform to promote their store or products.

Promotion options include:

- **Store Boost** – increases store visibility in search results
- **Product Boost** – highlights specific items
- **Featured Merchant Placement**

Payments for promotions are made via **M-Pesa**.

This feature is a **primary revenue source for the platform owner**.

4.3 Service Provider Module

Service providers operate similarly to merchants but focus on services rather than physical goods.

They can:

- create service listings
- define service areas
- set pricing ranges
- upload portfolio examples
- accept service requests
- receive ratings and reviews

Unlike product transactions, services typically do not require escrow unless prepaid booking is enabled.

4.4 Escrow Payment System

The escrow system is designed to ensure secure financial transactions.

Escrow Workflow

1. Buyer places order and pays via M-Pesa.
2. Payment is stored in the platform's escrow account.
3. Merchant confirms the order.
4. Platform delivery system picks up the product.
5. Item is delivered to the buyer.
6. Buyer confirms successful delivery.
7. Escrow releases payment to merchant.

If the buyer reports a problem, funds remain locked until a dispute is resolved.

4.5 Logistics and Delivery Module

The platform operates its own delivery network.

Delivery operations include:

- courier assignment
- pickup scheduling
- route optimization
- delivery tracking

The system ensures reliable delivery and provides updates to both buyers and merchants.

4.6 Reputation and Trust System

Trust is central to the platform.

Each merchant has a **Trust Score** calculated based on:

- customer ratings
- successful deliveries

- order completion rate
- customer complaints
- response time

Higher trust scores improve merchant ranking in search results.

4.7 Admin / Platform Owner Module

The platform administrator has full control over system operations.

Merchant Management

Admin can:

- review merchant applications
- verify business legitimacy
- create merchant accounts
- suspend or deactivate merchants

Product Moderation

Admin monitors marketplace listings and can remove:

- illegal products
- counterfeit goods
- prohibited items

Escrow Monitoring

Admin oversees all escrow transactions and can:

- release payments manually
- freeze suspicious transactions
- issue refunds

Dispute Resolution

Admin resolves conflicts between buyers and merchants by reviewing evidence and determining appropriate actions.

System Analytics

The admin dashboard provides insights such as:

- total transactions
- active users
- merchant growth
- delivery performance
- platform revenue

5. Revenue Model

The system owner generates revenue through multiple streams:

Merchant Promotion Fees

Merchants pay to **boost their stores or products** to gain higher visibility.

Transaction Commission

A small percentage is taken from each completed product sale.

Delivery Fees

Customers pay delivery charges for logistics services.

Featured Merchant Listings

Merchants can pay for premium placement on the platform.

6. Expected Impact

This platform addresses several major challenges in the Ethiopian commerce ecosystem:

- Lack of trust in online transactions
- Limited digital presence for local businesses
- Inefficient product discovery
- Unsafe payment practices
- Lack of organized delivery networks

By solving these challenges, the system has the potential to significantly improve **local commerce efficiency, merchant growth, and consumer confidence**.

7. Conclusion

The proposed platform represents a **comprehensive digital commerce ecosystem** tailored to the needs of the Ethiopian market. By integrating verified merchants, escrow-secured payments through M-Pesa, platform-managed logistics, and reputation-driven trust mechanisms, the system provides a secure and efficient environment for both buyers and sellers.

Beyond functioning as a marketplace, the platform serves as a **trusted infrastructure for modern local commerce**, empowering merchants to grow digitally while enabling buyers to transact safely and conveniently.